

Week 8 Workshop

Go through the script of two simple CNN models for Object detection and Semantic segmentation on FashionMNIST.

Object detection Architecture

```
Input: 1 × 28 × 28 (grayscale FashionMNIST image)
|
├─ Conv2d(1 → 32, kernel=3, padding=1) → ReLU
|   → Output: 32 × 28 × 28
├─ MaxPool2d(2) → Output: 32 × 14 × 14
|
├─ Conv2d(32 → 64, kernel=3, padding=1) → ReLU
|   → Output: 64 × 14 × 14
├─ MaxPool2d(2) → Output: 64 × 7 × 7
|
├─ Flatten → (64×7×7 = 3136)
├─ Linear(3136 → 128) → ReLU
├─ Linear(128 → 4)
|
└─ Sigmoid() → Bounding box coordinates ∈ [0,1]

Output: (B, 4)
```

Semantic segmentation Architecture

```
Input: 1 × 28 × 28
|
├─ Encoder
|   └─ Conv2d(1 → 16, kernel=3, padding=1) → ReLU → Output: 16 × 28 × 28
|       └─ MaxPool2d(2) → 16 × 14 × 14
|           └─ Conv2d(16 → 32, kernel=3, padding=1) → ReLU → 32 × 14 × 14
|               └─ MaxPool2d(2) → 32 × 7 × 7
|
├─ Decoder
|   └─ ConvTranspose2d(32 → 16, kernel=2, stride=2) → 16 × 14 × 14
|       └─ Conv2d(16 → 16, kernel=3, padding=1) → ReLU
|           └─ ConvTranspose2d(16 → 16, kernel=2, stride=2) → 16 × 28 × 28
|               └─ Conv2d(16 → 16, kernel=3, padding=1) → ReLU
|                   └─ Conv2d(16 → 1, kernel=1) → logits (no activation)
|
└─ Sigmoid (during inference)

Output: 1 × 28 × 28 (foreground probability map)
```